Traffic Light Controller using Verilog

VLSI Internship – Project Report

1. Aim

To design and simulate a Traffic Light Controller using Verilog HDL based on a Finite State Machine (FSM) and verify its operation using Xilinx Vivado.

2. Objective

* Design an FSM-based Traffic Light Controller.

* Simulate the controller.

* Verify correct traffic signal transitions.

* Perform RTL synthesis.

* Analyze synthesized hardware.

3. Software Used

* Xilinx Vivado

* Verilog HDL

4. Theory

Traffic management systems use controllers to regulate vehicle movement at road intersections. A Finite State Machine (FSM) provides a reliable method for controlling traffic lights by defining a sequence of states and transitions.

The controller cycles through four states:

1. North-South Green

2. North-South Yellow

3. East-West Green

4. East-West Yellow

Each state remains active for a specified number of clock cycles before transitioning to the next state.

5. Inputs

* Clock (clk)

* Reset (reset)

6. Outputs

* NS_R

* NS_Y

* NS_G

* EW_R

* EW_Y

* EW_G

7. Design Methodology

1. Defined FSM states.

2. Implemented state transition logic.

3. Designed output logic.

4. Developed a Verilog testbench.

5. Simulated the design.

6. Verified waveforms.

7. Performed RTL synthesis.

8. State Transition Table

Current State	Next State
S0	S1
S1	S2
S2	S3
S3	S0

9. Simulation Results

The simulation verified:

* Correct state transitions.

* Proper traffic light sequencing.

* Successful reset operation.

* Continuous cyclic operation.

Waveforms confirmed that the controller functioned as intended.

10. Synthesis Results

RTL synthesis completed successfully using Vivado. The synthesized design accurately represented the FSM-based traffic controller with minimal hardware resource utilization.

11. Applications

* Smart traffic intersections

* Urban traffic management

* FPGA learning projects

* Digital logic laboratories

* Intelligent transportation systems

12. Learning Outcomes

* Verilog HDL programming

* Finite State Machine design

* RTL development

* Testbench creation

* Simulation and debugging

* Logic synthesis using Vivado

13. Conclusion

The Traffic Light Controller was successfully designed, simulated, and synthesized using Verilog HDL in Xilinx Vivado. The project demonstrated the practical implementation of FSM concepts and strengthened understanding of the complete RTL design flow. It serves as a solid foundation for more advanced VLSI projects involving communication protocols, memory systems, and processor design.